

PHASE 3 – PROJECT DESIGN

1. INTRODUCTION

The Project Design phase defines the structure and working model of the **Script-Controlled ACL – Restrict Record Access Based on Field Value** project.

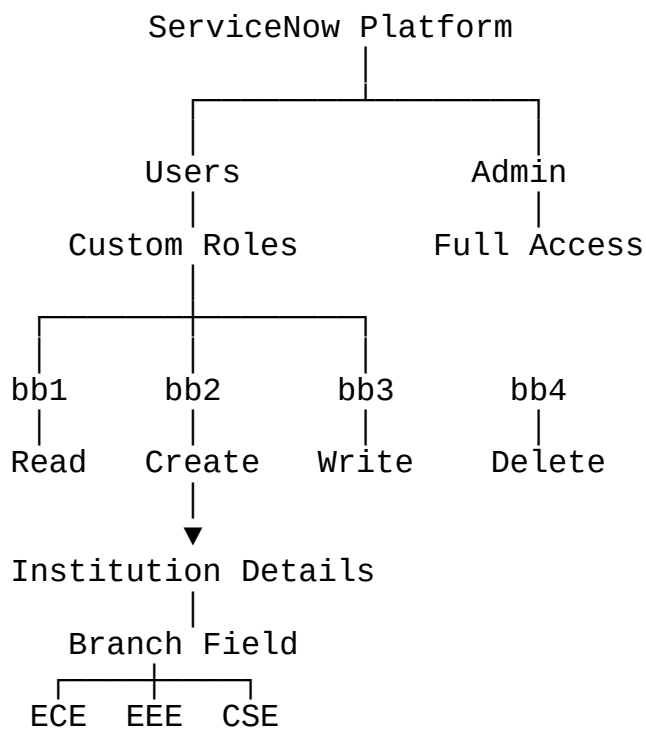
The design explains how users, roles, the Institution Details table, and ACLs work together to control access to records.

2. SYSTEM ARCHITECTURE

The proposed system consists of the following components:

- ServiceNow Platform
- Users
- Custom Roles
- Institution Details Table
- Read, Create, Write, and Delete ACLs
- Branch-Based Access Condition

Basic Architecture



3. TABLE DESIGN

The main table used in the project is:

Table Label: Institution Details

Table Name: u_institution_details

Field	Type	Purpose
Student Roll Number	Auto Number	Unique record identification
Student Name	Reference – User	Student information
Faculty Name	Reference – User	Faculty information
Branch	Choice	ECE, EEE, CSE
Email	String	Email details
Phone Number	String	Contact details
Description	Multi-line String	Additional information

The **Branch** field is the main field used for the access-control condition.

4. ROLE DESIGN

Four custom roles are designed for controlling different operations.

Role	Permission
bb1	Read
bb2	Create
bb3	Write
bb4	Delete

The roles can be combined to provide additional permissions.

5. ACL DESIGN

Separate ACLs will be created for each operation.

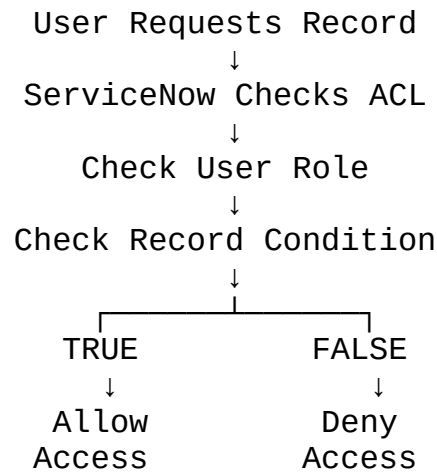
ACL	Operation	Role	Condition
Read ACL	Read	bb1	Branch is EEE
Create ACL	Create	bb2	No specific condition
Write ACL	Write	bb3	No specific condition
Delete ACL	Delete	bb4	No specific condition

The Read ACL uses an advanced script along with the Branch condition.

Administrators are designed to retain full access.

6. ACCESSFLOW

The basic access flow is:



For the Read operation, the system evaluates the user's authorization together with the configured **Branch = EEE** condition.

7. SECURITY DESIGN

The security design follows a combination of:

Role-Based Access + Record-Based Condition

For example:

- bb1 → Read permission
- bb2 → Create permission
- bb3 → Write permission
- bb4 → Delete permission
- admin → Full administrative access

This structure provides separate control over each operation.

8. Testing Design

The system will be tested using different role combinations.

Test User	Expected Access
User without required role	Restricted
User with bb1	Read permitted records
User with bb1 + bb2	Read + Create
User with bb1 + bb2 + bb3	Read + Create + Write
User with all four roles	Read + Create + Write + Delete
Administrator	Full access

Screenshots will be captured during implementation and testing as project evidence.

9. DESIGN OUTCOME

The Project Design phase establishes:

- The table structure.
- The role structure.
- The ACL structure.
- The Branch-based access condition.
- The user access flow.
- The testing approach.

This design will be used as the reference for the next phase, **Project Planning**, before starting the actual ServiceNow implementation.